Arkaprava Saha

Curriculum Vitae

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Webpage

Experience

July 2025 – Present **Assistant Professor**, *University of Tsukuba*, Tsukuba, Japan, Institute of Library, Information and Media Science.

- **Description:** Explaining neural networks and graph algorithms with human feedback. Currently a Visiting Researcher at Université Grenoble Alpes (UGA).

October 2024 – June 2025 **Postdoctoral Researcher**, *Université Grenoble Alpes (UGA)*, Grenoble, France, Laboratoire d'Informatique de Grenoble (LIG).

- **Supervisor:** Prof. Silviu Maniu
- **Description:** Explainability of data algorithms based on provenance

January 2023 – August 2024 **Research Fellow**, *CNRS@CREATE*, Singapore, DesCartes Program.

- **Supervisors:** Prof. Xiaokui Xiao (NUS), Prof. Bogdan Cautis (Univ. Paris-Saclay)
- **Description:** Learning from smart and complex data for Hybrid AI, primarily related to information propagation models and algorithms in graphs

May 2017 – July 2017 **Software Engineering Intern**, *Samsung R&D Institute*, Bangalore, India, Android Platform Team.

- **Mentor:** Preeti Rani Deka
- **Description:** Designed a generalized parser for any arbitrary watch face (dial) of the Samsung GWD to be rendered on an Android phone, thereby improving the old parser which was completely hardcoded for some specific watch faces only

Education

2018–2022 **Doctor of Philosophy (PhD) in Computer Science and Engineering (CSE)**, *Nanyang Technological University (NTU)*, Singapore, Data Management and Analytics Laboratory (DMAL).

- **Thesis:** Efficient Optimization of Network Metrics in Large Uncertain Graphs
- **Advisor:** Prof. Arijit Khan (till March 2022), Prof. Cheng Long (after March 2022)
- **Graduate Courses:** Information Retrieval and Analysis, Data Mining, Deep Learning for Data Science, Advanced Topics in Distributed Systems, Artificial Intelligence in Game Design, Database Systems
- **CGPA:** 4.0/5.0
- **Certifications**: Algorithms, Machine Learning, Deep Learning, Explainable AI

2014–2018 **Bachelor of Technology (BTech) in Computer Science and Engineering (CSE)**, *Indian Institute of Technology (IIT) Delhi*, New Delhi, India, Departmental Specialization in Data Analytics and Artificial Intelligence.

- **Thesis:** Random Walks and Reachability in Large Bloom Graphs
- **Advisor:** Prof. Maya Ramanath
- **Graduate Courses:** Machine Learning, Logic For Computer Science, Natural Language Processing, Numerical Algorithms
- **CGPA:** 8.187/10.0, **DGPA:** 8.223/10.0

Research Papers

For a quick overview, refer to my Google Scholar, DBLP and ResearchGate profiles. Note that in many research areas within computer science, including databases, data mining and artificial intelligence, conferences (not journals) are the primary venues for peer-reviewed publications.

Peer-reviewed Conference Papers

2025 **Hyperparametric Robust and Dynamic Influence Maximization**. Arkaprava Saha, Bogdan Cautis, Xiaokui Xiao, Laks V.S. Lakshmanan. *AAAI Conference on Artificial Intelligence (**AAAI**) 2025.*

2023 **Voting-based Opinion Maximization**. Arkaprava Saha, Xiangyu Ke, Arijit Khan, Laks V.S. Lakshmanan. *IEEE International Conference on Data Engineering (**ICDE**) 2023.*

2023 **Most Probable Densest Subgraphs**. Arkaprava Saha, Xiangyu Ke, Arijit Khan, Cheng Long. *IEEE International Conference on Data Engineering (**ICDE**) 2023.*

2021 **Shortest Paths and Centrality in Uncertain Networks**. Arkaprava Saha, Ruben Brokkelkamp, Yllka Velaj, Arijit Khan, Francesco Bonchi. *International Conference on Very Large Data Bases (**VLDB**) 2021.*

Peer-reviewed Journal Papers

2025 **Beyond Influence: Voting Theory for Opinion Maximization**. Arkaprava Saha, Xiangyu Ke, Arijit Khan, Laks V.S. Lakshmanan. *International Journal on Very Large Data Bases (**VLDB Journal**) 34(5), 57 (2025).*

Peer-reviewed Workshop Papers

2025 **Extending Layer-wise Relevance Propagation in Neural Networks using Semiring Annotations**. Antoine Groudiev, Arkaprava Saha, Silviu Maniu. *Provenance Week @ **SIGMOD** 2025.*

2019 **Reachability in Large Graphs using Bloom Filters**. Arkaprava Saha, Neha Sengupta, Maya Ramanath. *International Workshop on Large Scale Graph Data Analytics (LSGDA) @ **ICDE** 2019.*

Research Interests

I am currently working on algorithms for large graph data and explainable artificial intelligence (XAI), along with incorporating human feedback into them. Earlier, during and just after my PhD, my research focused on efficient and effective algorithms for large-scale uncertain graph data management and mining.

Teaching

Instructor

Tsukuba Frontiers of Informatics Research (2025-26 Semester 2)

Teaching Assistant (TA)

NTU Introduction to Databases (2018-19 Semester 1, 2019-20 Semesters 1 & 2, 2020-21 Semester 1)

NTU Data Structures and Algorithms (2018-19 Semester 2, 2020-21 Semester 2)

Supervision

PhD Students

Tsukuba Farnoosh Javar (co-supervised with Prof. Kei Wakabayashi)

Services

Reviewer or Program Committee (PC) Member

2027 VLDB

2026 SIGMOD, VLDB, AAAI, WSDM, CIKM

2025 WSDM, CIKM

2024 KDD, ACML, EAI MTYMEX

2023 ICDE (TKDE Poster Track)

2022 EAI ICECI

2021 SEA Data @ VLDB

Journal Reviewer

2025 Knowledge and Information Systems

2024 La Matematica, INFORMS Journal on Computing

2023 The VLDB Journal

External Reviewer

2025 VLDB

2023 ICDE

2022 VLDB, SIGMOD, KDD

2021 ICDE, VLDB

2020 SIGMOD

2019 VLDB, WWW, CIKM, DSAA

Volunteering

2022 Interview with Prof. Hanwang Zhang, winner of the Singapore Young Scientist Award 2021

2020 VLDB Student Volunteer

Positions of Responsibility

2019-2020 President of NTU School of CSE Graduate Students' Club

Talks

March 2026 Efficient Optimization of Network Metrics in Large Uncertain Graphs. *University of Tsukuba, Tsukuba, Japan.*

May 2024 Efficient Optimization of Network Metrics in Large Uncertain Graphs. *INRIA, Lille, France.*

July 2023 Efficient Optimization of Network Metrics in Large Uncertain Graphs. *University of Sydney, Sydney, Australia.*

July 2023 Efficient Optimization of Network Metrics in Large Uncertain Graphs. *RMIT University, Melbourne, Australia.*

March 2023 Most Probable Densest Subgraphs. *CNRS@CREATE, Singapore.*

March 2023 Voting-based Opinion Maximization. *CNRS@CREATE, Singapore.*

Technical Skills

Languages C, C++, Python, Java, SQL, SML

Software MATLAB, LaTeX, Android Studio

Honors and Awards

2024 **AI4Science and Nobel Turing Challenge Initiative (NTCI) Conference**, *AI for Science Initiative*, Singapore.
Won the best poster award among 80 posters selected for the conference

2024 **Global Young Scientists Summit (GYSS)**, *National Research Foundation (NRF)*, Singapore.
Among 350 young researchers (worldwide) selected to attend the event featuring winners of the Nobel Prize, Turing Award etc., and among 120 of the above 350 researchers selected to present research posters in the event

2021 **VLDB Conference Attendance Support Award**, *Very Large Data Base (VLDB) Endowment*, United States of America.
Awarded the grant for attending the conference

2019 **University Teaching for Teaching Assistants**, *Nanyang Technological University (NTU)*, Singapore.
Officially certified by the university as "Recommended to Teach"

2018 **Commonwealth Research Scholarship**, *Nanyang Technological University (NTU)*, Singapore.
Awarded the scholarship for pursuing PhD

2014 **JEE Advanced (IIT-JEE)**, *Indian Institutes of Technology (IITs)*, India.
Secured **All India Rank 98** out of around 1.3 million students and **State Rank 1**

2014 **West Bengal Joint Entrance Examination (WBJEEM)**, *West Bengal Joint Entrance Examinations Board (WBJEEB)*, India.
Secured **State Rank 6**

2013 **Kishore Vaigyanik Protsahan Yojana (KVPY)**, *Department of Science and Technology (DST)*, Government of India.
Selected as a **Fellow** in the SA stream with **All India Rank 56** out of around 0.15 million candidates

2012 **Regional Mathematical Olympiad (RMO)**, *Homi Bhabha Centre for Science Education (HBCSE)*, India.
Among 35 students in West Bengal to qualify for the national level examination

2010 **National Talent Search Examination (NTSE)**, *National Council of Educational Research and Training (NCERT)*, India.
Selected as a **Scholar**

Languages

Bengali	*Mother Tongue*
English	*Academic and Professional*
Hindi	*Advanced Conversations*
French	*Basic Conversations*